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The Size of the Middle Temporal Area in Primates

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With 7 Figures and 3 Tables

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Summary: The increase in the size of the lateral geniculate body and the primary visual cortex from prosimians to apes and man reflects at an anatomical level the importance of the visual system in primates. In the prestriate cortex visual modalities are processed separately in specialized pathways and areas. This arrangement facilitates the allometric analysis of cortical areas subserving defined visual modalities.

Its heavy myelination makes the middle temporal area, a visual cortical field specialized for the detection of moving stimuli, an easily detectable and reliably delineable area in histological sections. The size and position of the middle temporal area can therefore be compared between species, in order to collect quantitative data about the development of a defined visual submodality during primate evolution.

The volume of the middle temporal area was measured in 27 primate species. Allometric comparisons show that the middle temporal area is larger in simians than in most prosimians. In Callitrichidae, both the middle temporal area and the striate cortex are well developed. In cebids and cercopithecids, however, the sizes of the middle temporal area and primary visual cortex show divergent trends. Whereas the striate cortex is still enlarging, the size of the middle temporal area is reduced as compared to callitrichids.

Previous studies have revealed a close correlation between area striata and neocortex sizes, as well as area striata and lateral geniculate sizes. Such a close correlation does not exist for the middle temporal area versus neocortex or area striata. Therefore, the size of a visual structure serving a special submodality (e.g., the middle temporal area for the detection of moving stimuli) may develop in a species relatively independently from the lateral geniculate and primary visual cortex sizes.

Key words: Middle temporal visual area, primary visual cortex, lateral geniculate body, evolution, allometry

Introduction

The visual cortex of primates can be subdivided into numerous anatomically and functionally differing areas. It has been shown in allometric studies that the primary visual (striate) cortex enlarges considerably from prosimians to simians and also within the latter group from Callitrichidae to Cebidae and Cercopithecidae (FRAHM et al., 1984). This development reflects the increasing importance of the visual system during primate brain evolution in general.

Information from the primary visual cortex is processed further in more rostrally situated, specialized visual areas, representing different stations in parallel bundles of fiber tracts originating in the primary visual cortex (ZEKI, 1990; VAN ESSEN, 1985). One of these areas with well known function is the middle temporal visual area (MT), which contains many neurons sensitive to moving stimuli in the visual field.

WOOLSEY et al. (1955) described this cortical field, which is located in front of the circumstriate belt, in

the new world monkey *Callithrix*. ALLMAN and KAAS (1971) observed a comparable cortical area in *Aotus* and named it, according to its position in the temporal lobe, the "Middle Temporal Visual Area". The main anatomical criterion for the identification of this area is its very high density of myelin, which contrasts with lower densities in the surrounding cortical areas. In *Macaca*, for example, UNGERLEIDER and DESIMONE (1986a,b) have demonstrated that the area immediately lateral to MT, V4t, is characterized by light myelination in all cortical layers. The cortex medial to MT in the floor and the upper bank of the superior temporal sulcus is also only lightly myelinated, followed, further dorsally, by a second densely myelinated zone (DMZ). Another criterion for the delineation of MT is the decreased cell packing density in layer V (SPATZ and TIGGES, 1972).

From single cell recordings it became evident that neurons in the MT showed a stronger response to moving than to stationary stimuli: 60% responded exclusively to objects moving in one direction, and 24% responded best to movement in one direction

Table 1. Body- and brain-weights and volumes of visual brain structures in primates

Species	BoW (g)	BrW (mg)	AS (mm ³)	MT (mm ³)	CGL (mm ³)
<i>Microcebus murinus</i>	54	1780	136	13.2	7.1
<i>Cheirogaleus major</i>	450	6800	433	52.1	21.5
<i>Indri indri</i>	6250	38300	1648	161.6	94.5
<i>Loris tardigradus</i>	322	6600	529	32.9	24.6
<i>Galago senegalensis</i>	186	4800	301	39.8	16.0
<i>Otolemur crassicaudatus</i>	850	10300	479	47.5	23.2
<i>Galagoides demidoff</i> (n=2)	81	3380	214	19.7	10.9
<i>Tarsius spec.</i>	125	3600	349	49.7	20.8
<i>Callithrix jacchus</i> (n=3)	280	7600	566	68.2	25.1
<i>Cebuella pygmaea</i>	120	4150	363	41.9	16.9
<i>Saguinus midas</i>	340	10300	841	91.0	36.0
<i>Saguinus oedipus</i>	380	10000	807	103.1	34.2
<i>Callimico goeldi</i>	480	11000	987	113.7	31.7
<i>Cebus spec.</i>	3100	71000	3764	152.4	137.0
<i>Aotus trivirgatus</i>	830	17100	996	99.5	32.9
<i>Callicebus moloch</i> (n=2)	900	19000	1201	92.1	54.2
<i>Pithecia monachus</i>	1500	35000	1679	103.2	76.3
<i>Alouatta seniculus</i>	6400	52000	1918	220.0	87.4
<i>Macaca mulatta</i>	7800	93000	5343	332.0	158.0
<i>Cercocebus albigena</i>	7900	104000	5420	362.7	182.0
<i>Papio anubis</i>	25000	201000	10001	1052.6	395.0
<i>Cercopithecus ascanius</i>	3400	67000	4272	237.1	147.0
<i>Cercopithecus mitis</i>	6300	75000	4232	242.8	150.0
<i>Miopithecus talapoin</i>	1200	40000	2339	136.4	109.0
<i>Erythrocebus patas</i>	7800	108000	7775	335.4	266.8
<i>Nasalis larvatus</i>	14000	97000	4609	364.9	156.6
<i>Pan troglodytes</i>	46000	405000	10593	1567.1	356.0

AS = Area striata volume; most data taken from FRAHM et al., 1984

BoW = Body weight; data taken from STEPHAN et al., 1988

BrW = Brain weight; data taken from STEPHAN et al., 1988

CGL = Corpus geniculatum laterale volume; most data taken from STEPHAN et al., 1984

MT = Middle temporal visual area volume

bold face = new measurements

with a weaker response to the opposite direction (ALBRIGHT et al., 1984; *Macaca fascicularis*). The MT can therefore be classified as a field specialized for the analysis of the direction in which stimuli move in the visual field. The MT corresponds to Area V 5 in ZEKI's studies (1978, 1990) of the macaque visual cortex. Until now, its existence has been demonstrated in the simians *Callithrix*, *Aotus*, *Saimiri* and two species of *Macaca*, as well as in the prosimian *Galago* (ALLMAN and MCGUINNESS, 1988).

The high myelin density of the MT facilitates reliable identification of the borders of this functionally defined area. Thus, a comparative neuroanato-

mical study on the size of area MT in primates becomes possible when rare prosimian species, which are crucial for a comparative allometric study, are included. The present investigation provides information on the size of the MT in prosimians and simians and the concomitant evolution of the primary visual cortex.

Material and Methods

The volume of the MT was measured in 8 prosimian species and in 19 species of simians (see Table 1) by using serially sec-

Table 2. Percentages and ratios of Area MT

Species	MT in % of NEO	AS MT	MT CGL
<i>Microcebus murinus</i>	1.79	10.29	1.87
<i>Cheirogaleus major</i>	1.77	8.31	2.42
<i>Indri indri</i>	0.80	10.20	1.71
<i>Loris tardigradus</i>	0.93	16.09	1.34
<i>Galago senegalensis</i>	1.86	7.55	2.49
<i>Otolemur crassicaudatus</i>	1.01	10.09	2.05
<i>Galagoides demidoff</i> (n=2)	1.26	10.84	1.81
<i>Tarsius spec.</i>	2.81	7.02	2.39
(n=8)	1.53	10.05	2.01
<i>Callithrix jacchus</i> (n=3)	1.56	8.30	2.72
<i>Cebuella gygmaea</i>	1.79	8.67	2.48
<i>Saguinus midas</i>	1.55	9.24	2.53
<i>Saguinus oedipus</i>	1.75	7.83	3.01
<i>Callimico goeldi</i>	1.75	8.69	3.58
(n=5)	1.68	8.55	2.86
<i>Cebus spec.</i>	0.33	24.69	1.11
<i>Aotus trivirgatus</i>	1.09	9.21	3.29
<i>Callicebus moloch</i> (n=2)	0.83	13.03	1.70
<i>Pithecia monachus</i>	0.49	16.27	1.35
<i>Alouatta seniculus</i>	0.69	8.72	2.52
(n=5)	0.69	14.38	1.99
<i>Macaca mulatta</i>	0.52	16.09	2.10
<i>Cercocebus albigena</i>	0.52	14.94	1.99
<i>Papio anubis</i>	0.75	9.50	2.66
<i>Cercopithecus ascanius</i>	0.52	18.02	1.61
<i>Cercopithecus mitis</i>	0.49	17.43	1.62
<i>Miopithecus talapoin</i>	0.52	17.15	1.25
<i>Erythrocebus patas</i>	0.43	23.18	1.26
<i>Nasalis larvatus</i>	0.58	12.63	2.33
(n=8)	0.54	16.12	1.85
<i>Pan troglodytes</i>	0.54	6.76	4.40

NEO = Neocortex volume; data taken from FRAHM et al., 1982

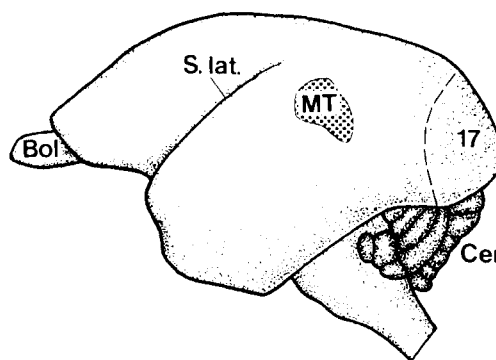


Fig. 1. Lateral view of the brain of the common marmoset, *Callithrix jacchus*, indicating the position and extent of Area MT and the primary visual cortex (Area 17).

the size of the brain as a whole), the entire rostro-caudal extent of the MT. These values were corrected for shrinkage due to histological processing (STEPHAN et al., 1991). The resulting volumes of the MT in the 27 species were compared by applying allometric methods. The logarithmic expression of the allometric formula (SNELL, 1892)

$$\log y = \log b + a \cdot \log x$$

represents a linear regression (y = MT volume, b = intersection of the regression line with the y -axis, x = body weight), in which the slope a of the allometric line was determined by a principal component analysis within both prosimian and simian data. A detailed description of this mathematical procedure is given in REHKÄMPER et al. (1991). The principal component analysis allows the calculation of a common slope ($a = 0.513$) for the allometric line with contributions (weighted by the sample size) from both the prosimian and simian groups. After the calculation of this common slope, the regression line was positioned so as to run through the average body weight and average MT volume of the 8 prosimian species. The deviation of each single primate species from this baseline was calculated as size index (see Figs. 6, 7, and Table 3) indicating the increase or decrease of the respective area.

Results

In histological sections stained for myelinated fibers, area MT appears as a field characterized by a dense band of darkly staining fibers, mainly occupying the lower cortical layers. In smaller, almost lissencephalic prosimian and simian brains this cortical area is situated on the dorsolateral surface of the temporal lobe. Figure 1 shows the position and extent of the MT in the brain of the common marmoset, *Callithrix jacchus*.

To illustrate the position of the MT in another almost lissencephalic callitrichid species, a sequence of schematic drawings of coronal sections through the brain of *Saguinus oedipus* is given in Figure 2. The section numbers indicate that the rostro-caudal

tioned, Nissl- and myelin-stained brains from Stephan's and our own collections, both located at the C. & O. Vogt Institute of Brain Research.

Additional information for a reliable delineation came from receptor-autoradiographic material and corresponding sections stained for myelinated fibers in brains of *Macaca mulatta*.

Animals used for histological processing of brains were sacrificed by a lethal dosis of sodium pentobarbital and perfused transcardially with physiological saline followed by Bouin's fluid or Bodian's fixative. The brains were carefully dissected, embedded in paraffin, cut in the coronal plane and cell bodies or myelinated fibers were stained in alternating sections (for details see STEPHAN et al., 1981).

Volumes were determined by planimetry of camera lucida drawings covering, at regular distances (assessed according to

Table 3. Allometric size indices of visual structures in primates

	MT	AS	CGL
	a = 0.513	a = 0.510	a = 0.480
	b = 0.350	b = 1.277	b = 0.074
Prosimians			
<i>Microcebus murinus</i>	76	94	88
<i>Cheirogaleus major</i>	101	101	97
<i>Cheirogaleus medius</i>	-	75	76
<i>Lemur albifrons</i>	-	165	135
<i>Varecia variegata</i>	-	155	125
<i>Lepilemur ruficaudatus</i>	-	53	50
<i>Avahi laniger laniger</i>	-	65	75
<i>Avahi l. occidentalis</i>	-	76	96
<i>Propithecus verreauxi</i>	-	91	109
<i>Indri indri</i>	81	101	120
<i>Daubentonia</i>			
<i>madagascariensis</i>	-	103	109
<i>Loris tardigradus</i>	76	147	130
<i>Nycticebus coucang</i>	-	122	122
<i>Perodicticus potto</i>	-	72	69
<i>Galago senegalensis</i>	122	111	110
<i>Otolemur crassicaudatus</i>	66	81	77
<i>Galagoides demidoff</i>	92	120	111
<i>Tarsius spec.</i>	186	157	173
Simians			
<i>Callithrix jacchus</i>	169	169	141
<i>Cebuella pygmaea</i>	160	155	133
<i>Saguinus midas</i>	204	228	185
<i>Saguinus oedipus</i>	218	206	166
<i>Callimico goeldi</i>	213	224	138
<i>Cebus spec.</i>	110	330	244
<i>Aotus trivirgatus</i>	141	171	110
<i>Callicebus moloch</i>	125	198	175
<i>Saimiri sciureus</i>	-	363	235
<i>Pithecia monachus</i>	108	213	192
<i>Alouatta seniculus</i>	109	116	110
<i>Ateles geoffroyi</i>	-	203	170
<i>Lagothrix lagothricha</i>	-	327	228
<i>Macaca mulatta</i>	149	292	181
<i>Cercopithecus albigena</i>	162	295	207
<i>Papio anubis</i>	260	302	258
<i>Cercopithecus ascanius</i>	163	357	250
<i>Cercopithecus mitis</i>	122	258	190
<i>Miopithecus talapoin</i>	160	333	307
<i>Erythrocebus patas</i>	150	425	305
<i>Colobus badius</i>	-	177	154
<i>Nasalis larvatus</i>	121	187	135
<i>Hylobates lar</i>	-	294	233
<i>Pan troglodytes</i>	283	235	173
<i>Gorilla gorilla</i>	-	175	126
<i>Homo sapiens</i>	-	274	172

a = slope of the allometric line; b = intersection with the y-axis

Table 3A. Summary of Table 3

	n	MT	n	AS	n	CGL
Prosimians	8	100	18	105	18	104
Non-human simians	19	165	25	245	25	190
marmosets	5	193	5	190	5	156
monkeys	13	145	17	262	17	203
apes	1	283	3	235	3	177
Man	-	-	1	274	1	172

Indices of Area striata (AS) taken from FRAHM et al., 1984

Indices of Corpus geniculatum laterale (CGL) taken from STEPHAN et al., 1984

(n) = number of species

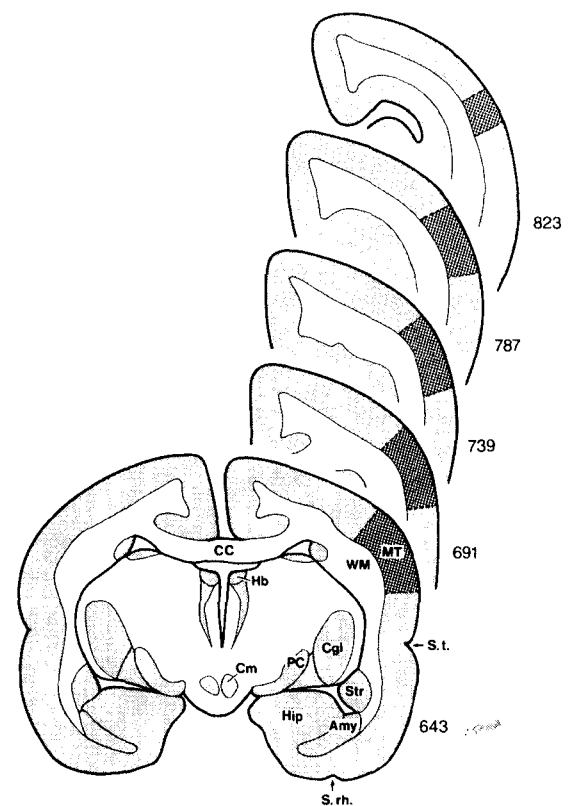


Fig. 2. Sequence of five coronal sections (20 µm each) from rostral to caudal through the brain of *Saguinus oedipus* 1220. Numerals indicate section numbers.

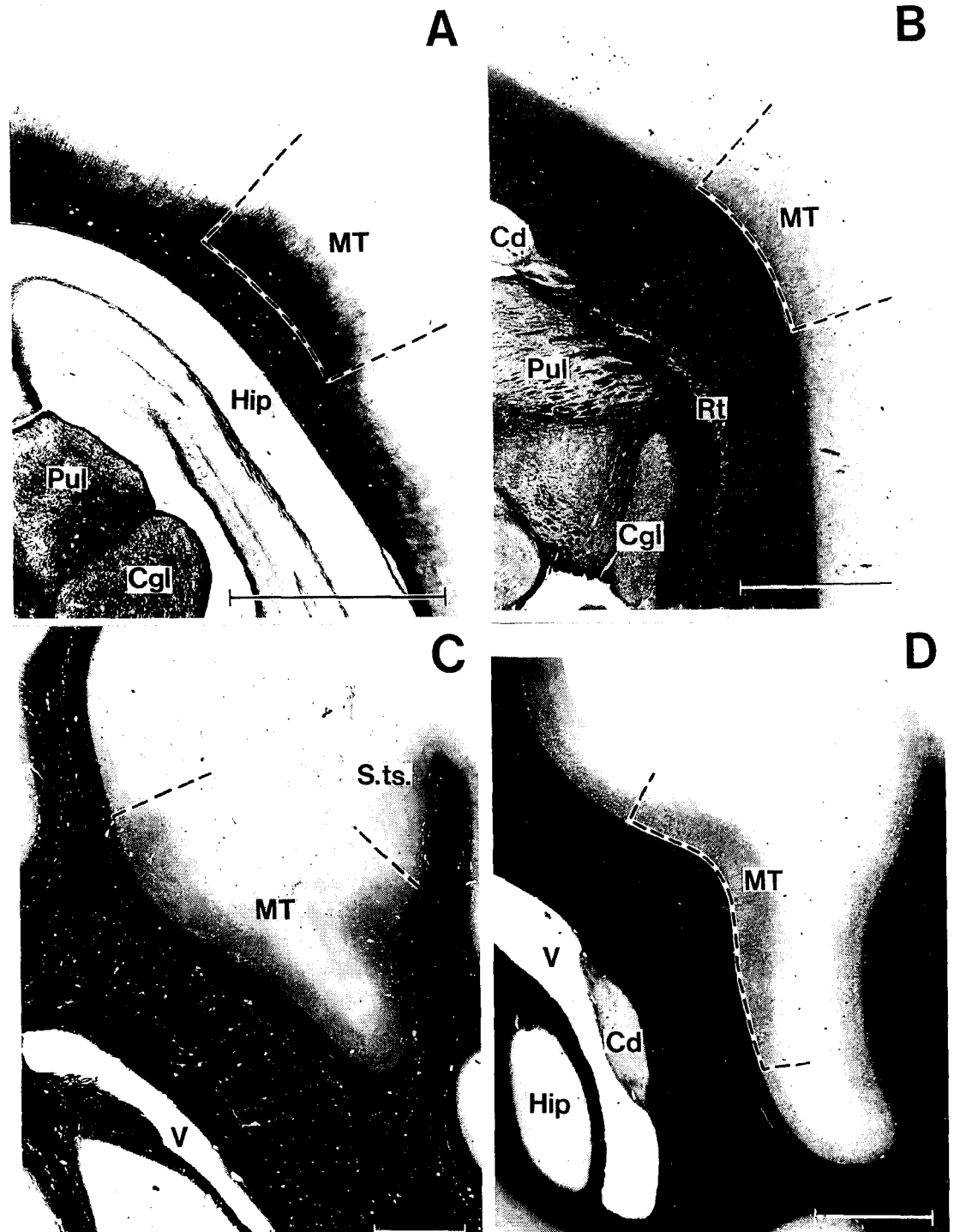


Fig. 3. Coronal sections through the brain at the level of Area MT.

A) *Galagoides demidoff* A 281, Gallyas stain, 20 μ m; B) *Saguinus midas tamarin* 1182, Heidenhain-Woelcke stain, 20 μ m; C) *Cercopithecus ascanius* A 219, Heidenhain-Woelcke stain, 20 μ m; D) *Miopithecus talapoin* 1201, Heidenhain-Woelcke stain, 20 μ m; Scale bar: 2 mm.

extent of the MT is approximately 3.6 mm in this species.

In the convoluted brains of larger prosimians and simians, the MT is found close to the superior temporal sulcus. Quite often, it disappears from the sur-

face and is situated in the floor of this sulcus (e.g., *Cercopithecus ascanius* in Fig. 3C).

Figure 4 shows two adjacent horizontal sections through the brain of *Macaca mulatta*. The heavily myelinated MT stains darker than the surrounding

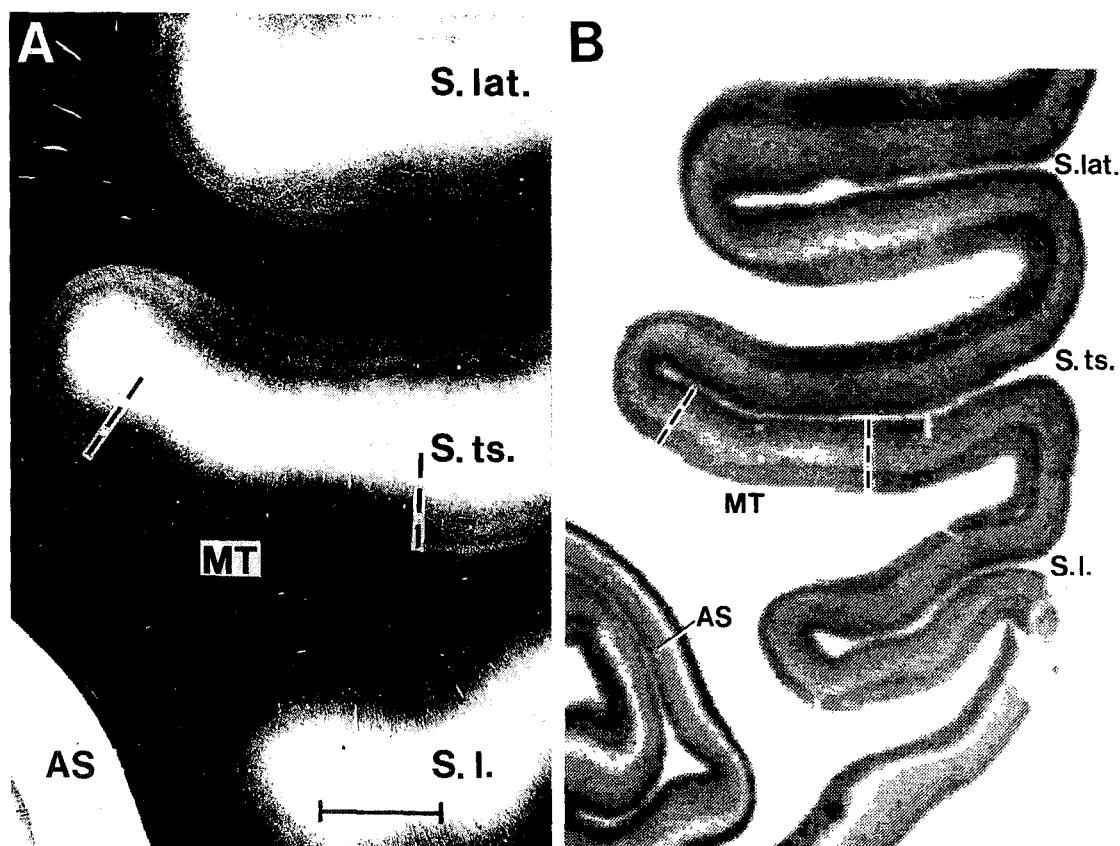


Fig. 4. Horizontal sections through the brain of *Macaca mulatta*.

A) Gallyas stain for myelinated fibers. Scale bar: 2 mm.

B) Plot of kainate receptor densities in an adjacent section.

Note the highly contrasted, striped appearance of cortex in the primary visual area (AS).

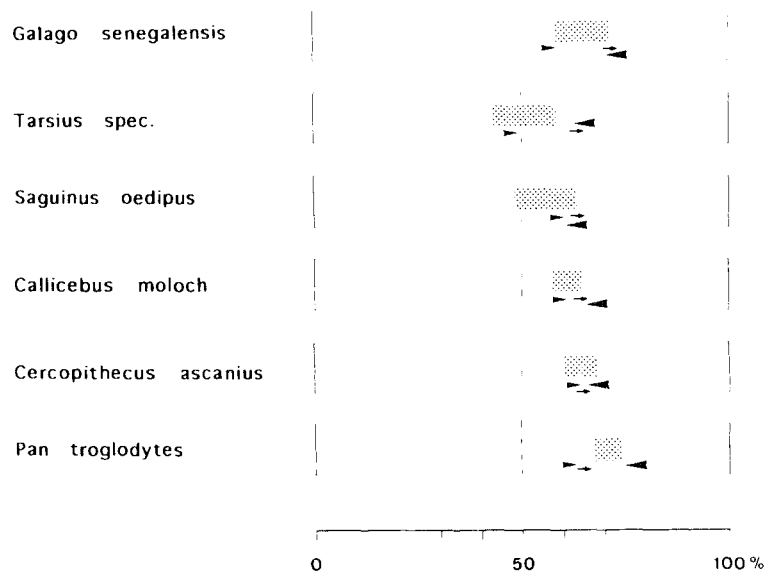


Fig. 5. Rostro-caudal extent of Area MT in relation to hemisphere length in six primate species.

Hemisphere length = 100%. Hatched area: Length and position of Area MT. ➤ = caudal end of caudate nucleus, ➡ = caudal end of hippocampus, ◀ = rostral beginning of Area striata. Position of 50 % of hemisphere length is marked by dotted lines.

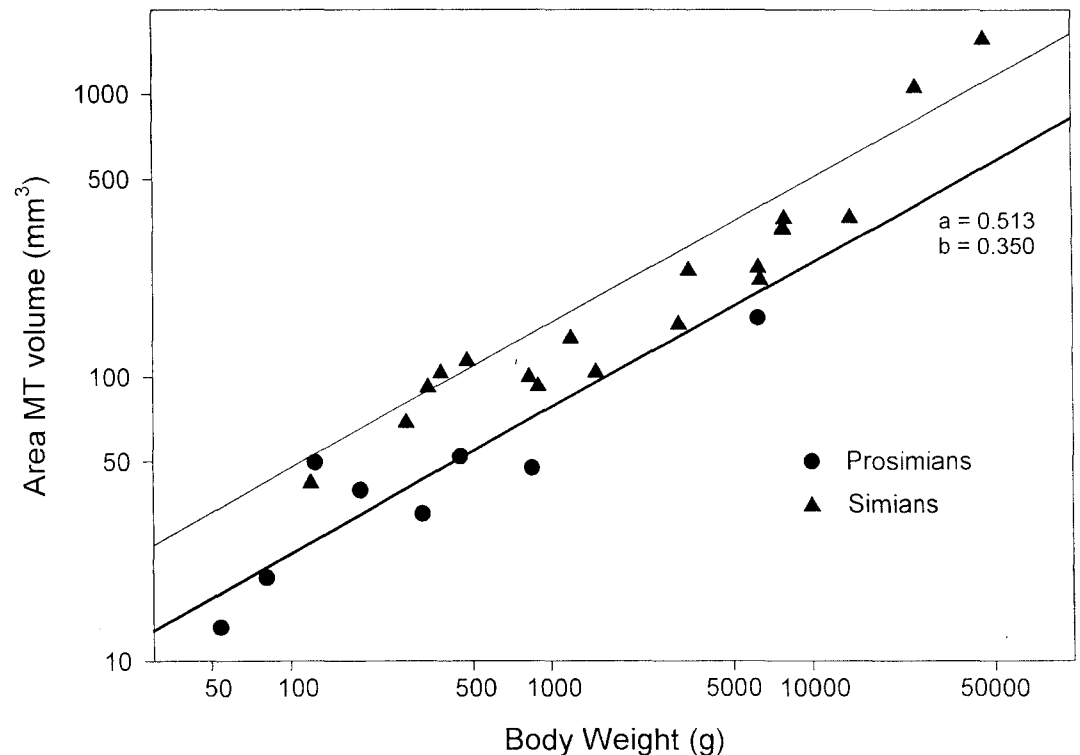


Fig. 6. Log-log plot of Area MT volume (in mm³) versus body weight (in g). The thick reference line running through the prosimian species has the equation: $\log \text{Area MT volume} = 0.350 + 0.513 \times \log \text{body weight}$. The thin parallel line indicates a twofold enlargement of Area MT size. The prosimian species lying close to the upper line is *Tarsius*.

fields. In the corresponding autoradiographic preparation, the density of kainate receptors (a glutamate receptor) is depicted. MT is characterized by a broad band of low receptor density in the middle cortical layers, and, additionally, a relatively low density in deep cortical layers close to the underlying white matter. Delineations based on myeloarchitecture and receptor-autoradiography coincide.

In Figure 5 the rostro-caudal extent of Area MT is seen in relation to the length of the cerebral hemispheres (= 100 %). For the sake of simplicity only one species from each group of primates was chosen for this graph. In prosimians and Callitrichidae, the MT covers a large area on the lateral surface of the temporal lobe. In all simian species other than Callitrichidae, the rostro-caudal extent of the MT seems to be much smaller in relation to the total length of the cerebral hemisphere. The position of the MT in *Tarsius* is remarkable. Its rostral pole lies at approximately 40 % of the length of the hemisphere, i.e., the most anterior location in any of our species. This position is due to the highly developed primary visual cortex in this nocturnal prosimian. The primary visual cortex of *Tarsius* occupies the

entire occipital third of the hemisphere and seems to push the MT into an unusual rostral position. In monkeys and apes, the opposite effect is observed. In these species, the frontal areas of the cerebral hemispheres show a highly progressive development. Thus, the MT is located in a more posterior position. The rostrocaudal extent of the MT coincides approximately with the levels at which the caudal poles of the caudate nucleus and hippocampus, and the rostral border of Area 17 (Fig. 5) on the medial surface of the hemisphere are found.

The relative volume of the MT, compared with that of the entire neocortex (= 100 %), is listed in Table 2. In prosimians, 1.5% of the neocortex is occupied by the MT. This value increases to 1.7% in Callitrichidae and then continually decreases in monkeys and apes.

The ratio of the volume of the primary visual cortex to that of the MT shows that the striate area is 10 times larger than the MT in prosimians. Among simians, Callitrichidae have a relatively large MT, resulting in Area striata/MT ratios below 10. In Cebidae and Cercopithecidae, on the other hand, this ratio is clearly above 10 in most species.

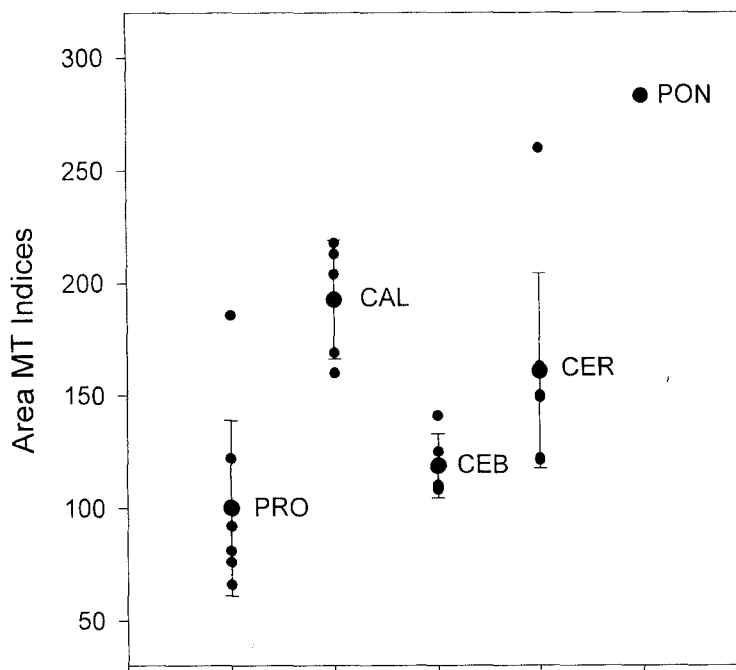


Fig. 7. Size indices of Area MT. Individual indices are marked by small black dots. Averages and standard deviations in taxonomic groups are given by big black dots and vertical lines, respectively.

The MT has a volume twice that of the lateral geniculate body (CGL) in prosimians, and similar ratios are found in Cebidae and Cercopithecidae, whereas in Callitrichidae (2.9) and Pongidae (4.4) the MT has an even larger ratio.

The size of the MT in prosimians was used as a reference for comparisons with simians (Fig. 6), i.e. the average MT index of the eight prosimian species is 100. The thinner parallel line in Figure 6 indicates a twofold enlargement of the MT in comparison to its average size in prosimians. Some species show even greater deviations from the mean values in prosimians. In Figure 7 and Table 3 these deviations from the reference line are scaled and listed. The MT index is smallest in *Otolemur crassicaudatus*, the African Greater Bushbaby. The index of *Tarsius* (186) is more than twice the average of the seven other prosimian species. Besides *Tarsius*, only two prosimian species, *Galago* and *Cheirogaleus*, reach values above 100.

In the Callitrichidae, the MT is large in all species. Among the Cebids, *Aotus*, the only nocturnal species, has a higher MT index than any other member of its family. The indices of the Cebids, however, lie clearly below those of the Callitrichidae.

In the Cercopithecidae, indices are higher than in the Cebidae, but span a rather large range of values with a maximum in the great baboon, *Papio anubis*. The highest MT index is found in the chimpanzee,

indicating a 2.8-fold enlargement of this area compared to its size in an average prosimian.

Discussion

Comparison with areal measurements of the MT in other studies

In this study, the MT is defined as the densely myelinated area on the temporal lobe or in the depth of the superior temporal sulcus. Because most of the material consisted of serial sections stained with the Heidenhain-Woelcke technique for myelinated fibers and was not specifically prepared for demonstrating the MT, staining was relatively weak in the cortex. UNGERLEIDER and DESIMONE (1986a) could show that in the macaque, the heavy myelination of MT decreases to some extent toward the periphery of this field (MTp), but both MT and MTp are the representation of V1 projections to the superior temporal cortex. According to these authors, inclusion of MTp in the MT measurements would add approximately 20 percent in size. The areas immediately adjacent to MT and MTp are remarkably less myelinated (UNGERLEIDER and DESIMONE, 1986a). Given the relatively weak Heidenhain-Woelcke stain of cortical fibers in our material, our measurements only

include the MT proper (and probably minor parts of MTp), but none of the smaller adjacent visual areas (e.g., the densely myelinated zone in the dorsal upper bank of the superior temporal sulcus), which these authors could delineate additionally in Gallyas-stained material. Their data include areal measurements of the MT field in several *Macaca mulatta* specimens, and we have used their findings to estimate the degree of correspondance between their and our delineations.

In three *Macaca mulatta* specimens, UNGERLEIDER and DESIMONE (1986a) reported values of 52 mm², 52 mm², and 82 mm² for the size of the MT. From the known spacing of electrodes, they estimated a linear shrinkage of 12% during perfusion and histological processing. Under the assumption that cortical thickness is approximately 2 mm, this would lead to a volume of the MT in both hemispheres in *M. mulatta* of 360 mm³ for the three animals together, or 305 mm³ when the highly deviating value is excluded. Our value (332 mm³) for both sides and corrected for shrinkage is exactly in the same range, indicating the reliability of the delineation of this cortical area.

Surface areas of visual cortical fields expressed as percentages of the total neocortical surface are given by KRUBITZER and KAAS (1990) for *Saimiri*, *Aotus*, *Callithrix* and *Galago senegalensis*. According to these authors, the MT occupies 2.2% in *Galago*, and 1.8% in *Callithrix* and in *Aotus*. Our corresponding values for volume measurements are 1.9%, 1.6%, and 1.1%, respectively. Although the present data give slightly smaller values in all three cases, the comparability of the definition of the MT is given in all three studies.

The statement by KRUBITZER and KAAS (1990) that the MT may be relatively larger in nocturnal than in diurnal primates (expressed as percentages of the total neocortex) is corroborated and extended in our study of a much larger sample of different species. *Indri indri*, the only diurnal prosimian of our study, has a lower relative value for the MT than the nocturnal prosimians on the average. In *Aotus*, the only nocturnal simian, the relative value is higher than in the diurnal cebid species.

In *Callithrix penicillata* (which, according to CORBET and HILL (1980) is synonymous to *C. jacchus*), PESSOA et al. (1992) found a ratio between the areas of the MT and the striate cortex of 8%. In their comparisons with data from the literature (for *Macaca*, *Cebus*, and *Aotus*), they postulated the same ratio for all primates. KRUBITZER and KAAS (1990), however, showed that in *Galago senegalensis* the MT has a size of 13% relative to the striate area, indicating that such a constancy does not exist.

From our volumetric data (Tab. 1) on the MT and the primary visual cortex, a wide range of ratios from 4% in *Cebus* and *Erythrocebus* to 15% in *Pan* can be calculated. A constancy in the relative sizes of the MT to the striate cortex, as postulated by PESSOA et al. (1992) therefore does not exist.

Comparison with volumetric measurements of other visual areas

Although prosimians do have good visual capacities, visual cues seem to be of increasing importance in higher monkeys. Measurements of area striata volumes (FRAHM et al., 1984) showed on the average a 2.5 fold increase of allometric size indices in simians as compared to prosimians.

The dorsal nucleus of the lateral geniculate body, on the other hand, increases 1.9 times in simians in comparison to its average size in prosimians (STEPHAN et al., 1984). These authors could show that there exists a strong correlation between area striata and corpus geniculatum laterale sizes in primates ($r^2 = 0.985$). Species with large visual structures (in relation to body size) were *Tarsius* among the prosimians and *Erythrocebus*, *Miopithecus*, *Papio*, *Cercopithecus ascanius*, *Saimiri* and *Lagothrix* among simian primates. Pongids and man did not show any further relative enlargement of visual structures compared to cebids and cercopithecids.

The differences in the size of visual structures between prosimians and simians may reflect the greater complexities of the habitat, home range, occupied forest stratum, and especially, the many visual signals relevant to simians, that are linked to a diurnal life in social groups.

The size of MT seems to be relatively independent of the neocortex size ($r^2 = 0.922$), in contrast to area striata size, which shows a much closer correlation with the neocortex ($r^2 = 0.981$). Therefore, the species with large area striata and corpus geniculatum laterale sizes (relative to body weight) are not necessarily to be found in similar positions for MT size. Among the species with large visual structures mentioned above, only *Tarsius* and *Papio* also show a relatively large area MT.

Besides these two species and *Pan*, which belong to very diverse primate groups, one family is exceptional in its large MT size, the marmosets. One could argue that detection of movements plays a predominant role for these small primate species, which on one hand rely on a certain amount of animal prey (such as insects) in their diet and, on the other hand, are threatened by many more enemies (e.g., diurnal birds of prey, snakes) than the large cebid and cercopithecoid monkeys.

As mentioned above, the thalamic relay nucleus and the primary visual cortex enlarge considerably in simians as compared to prosimians (1.9 fold for the CGL, and 2.5 fold for the AS), and the sizes of both are closely correlated. From the primary visual cortex the information is fed into different channels, serving specified visual submodalities. These submodalities may be of differing importance in the biology of the various primate species, making these areas to a certain degree independent of the overall size of the visual apparatus.

Abbreviations

Amy	Corpus amygdaloideum
AS	Area striata
Bol	Bulbus olfactorius
BoW	Body weight
BrW	Brain weight
CAL	Callitrichidae
CC	Corpus callosum
Cd	Nucleus caudatus
CEB	Cebidae
CER	Cercopithecidae
Cer	Cerebellum
Cm	Corpus mamillare
Cgl	Corpus geniculatum laterale
Hb	Nuclei habenulae
Hip	Hippocampus
MT	Middle temporal visual area
NEO	Neocortex
PC	Pedunculus cerebri
PON	Pongidae
PRO	Prosimians
Pul	Pulvinar
Rt	Nucleus reticularis thalami
S. lat.	Sulcus lateralis
S. rh.	Sulcus rhinalis
S. t.	Sulcus temporalis
S.ts.	Sulcus temporalis superior
Str	Corpus striatum
V	Ventricle
WM	White matter

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